

Sujith S

Applied Scientist | Machine Learning · NLP · Deep Learning · Large Language Models · Distributed Systems
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Summary

Published researcher and Computational Statistics student at VIT University (2022-2027) with applied experience in machine learning, NLP, deep learning, and large language model systems. Proficient in Python, PyTorch, TensorFlow, and Hugging Face Transformers; experienced building NLP pipelines, RAG architectures, and large-scale distributed data processing platforms on AWS using Apache Kafka and Apache Spark. Passionate about advancing the state of the art in NLP, NLU, speech and language technology, and human language understanding systems.

Education

Int.MSc Computational Statistics and Data Analytics

VIT University | 2022-2027

Coursework: Machine Learning, NLP, Deep Learning, Statistical Modeling, Data Structures & Algorithms, Distributed Systems

CS50 Programs

Harvard CS50 (Online)

Completed [CS50x](#), [CS50P](#), [CS50AI](#), [CS50SQL](#), and [CS50R](#) covering algorithms, AI, Python, and full-stack systems.

Skills

Languages: Python, Java, SQL, R, JavaScript, TypeScript

Machine Learning & AI: Deep Learning, Reinforcement Learning, NLP, NLU, LLMs, Transformer Architecture, Transfer Learning, Fine-Tuning, Model Optimization, Inference Optimization

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph, FAISS, ChromaDB, XGBoost, Sentence Transformers, NLTK

Distributed Systems & Data: Apache Kafka, Apache Spark, Delta Lake, Apache Airflow, PostgreSQL, Redis, MLflow

Cloud & Infrastructure: AWS (Lambda, S3, SageMaker, EKS, EventBridge), Docker, CI/CD Pipelines, Linux/Unix

Research & Publications

Stock Market Analysis and Prediction Using Machine Learning

2025

[El Ruha International Conference on Scientific Research](#), pp. 730-740. Built a unified ML framework applying LSTM, GRU, Transformer, ARIMA, XGBoost, LightGBM, Reinforcement Learning (DQN, PPO, A2C), and ensemble models for financial time-series forecasting; developed feature engineering pipelines and evaluated 21 models for stock price prediction on real market data. [\[GitHub\]](#)

Projects

Social Media Sentiment Lakehouse

[GitHub](#)

Python, Hugging Face Transformers, Apache Kafka, Spark Streaming, Delta Lake, NLTK, MLflow, PostgreSQL, Docker, AWS

- Designed a large-scale NLP analytics platform ingesting real-time social streams via Apache Kafka; implemented Hugging Face Transformer pipelines for sentiment classification, emotion detection, and topic modeling over distributed data with Spark Streaming and Delta Lake; tracked ML experiments with MLflow and deployed inference services on AWS.

Autonomous Research Agent (RAG + NLP Pipeline)

[GitHub](#)

Python, LangChain, LangGraph, Hugging Face Transformers, Sentence Transformers, FAISS, ChromaDB, FastAPI, PostgreSQL, Docker, AWS

- Architected a RAG system using LangChain, LangGraph, and Hugging Face Sentence Transformers for semantic chunking, vector embedding, and similarity search via FAISS and ChromaDB; implemented LLM-based query decomposition and multi-step NLU reasoning; automated evaluation pipelines and benchmarked NLP model accuracy and retrieval precision.

Recommendation Engine (Deep Learning + NLP)

[GitHub](#)

Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, FAISS, Sentence Transformers, FastAPI, PostgreSQL, Redis, Docker

- Engineered a hybrid deep learning recommendation system combining collaborative filtering, content-based filtering, and embedding-based vector similarity search via FAISS; built model training pipelines in PyTorch and TensorFlow with ensemble learning, online model updates, and optimized low-latency inference APIs for large-scale user datasets.

Work Experience

Founder

[Shoot the Breeze](#) |
2023 - Present

- Owned end-to-end ML system development: designed NLP-powered features, implemented model training and inference pipelines in Python, and deployed services on AWS with Docker and automated monitoring workflows.

App/Web Developer

[CropSky](#) | VIT-TBI Research & Innovation Hub |
2024 - Present

- Developed ML-powered backend systems and REST APIs on cloud infrastructure at a university-incubated startup; implemented automated data pipelines and collaborated with engineers to integrate ML model outputs into production applications.

Certifications & Achievements

- [Google Generative AI Leader](#)
- [Google Cloud Computing Foundations](#)
- [FreeCodeCamp Legacy Full Stack](#)
- [OCI 2025 Certified Developer Professional](#)
- AWS [Solutions Architect](#) and [Developer](#)
- [Software Engineer Certification - HackerRank](#)